Premium or discount? Depends who's asking.

Replicating and extending the IFS study (Bee Boileau, Laurence O'Brien and Ben Zaranko, 2022) on the public-private sector wage differential across different groups in 2021-22 and 2022-23

1. Key findings

In 2021-22, public sector employees earned 7.1% more per hour than those in the private sector workers, yet 1.3% less once controlling for sex, experience, region, age, quarterly changes and education. The IFS (2022) estimates of 7% and -0.7% both lie within the study's 95% confidence intervals, and this study's estimates both lie within the IFS's confidence intervals. In 2022-23, the raw differential fell to 3.8%, and the conditional fell to -3.4%.

By sex, the conditional public-private sector wage differentials were 1.4% and -4.9% for females and males respectively in 2021-22. These numbers were 2.2% and -4.6% respectively in the IFS study and so estimates lay in IFS confidence intervals and vice versa. Whilst for males the differential is statistically different from zero, the female differential is not statistically different from zero. In 2022-23, the figures fell to -1% and -6.6% for females and males respectively, and again, the number for females is not statistically different from zero.

By region, the 2021-22 differential ranged from 8.3% in Scotland to -10.2% in the South East. In most cases, the 2022-23 differential favoured public sector workers less than in 2021-22. However, in both years, only three of the twelve regions returned a differential statistically different from zero.

The differential varies by highest qualification, with degree holders facing the highest public sector discount. Those with degrees faced a 5.4% discount in 2021-22, with the gap widening in 2022-23, whilst higher degree and A levels holder only face a significant differential in 2022-23, at -4.4% and -3.3% respectively.

The conditional differential isn't robust to how education is controlled for in 2021-22, yet it is robust in 2022-23. The combining of the higher degree and degree categories into one for workers with a degree or higher took the conditional differential from -1.3% to -0.67%, far closer to the IFS estimate. The gap is not significantly negative when using this less granular measure in 2021-22, though it is in 2022-23.

The write up proceeds as follows. Section 2 discusses the study's motivations and questions, section 3 outlines the data and methodology used, section 4 talks through the results, section 5 discusses the study's limitations, and section 6 concludes.

2. Motivations and questions

The project's purpose is to investigate how the public-private sector wage differential changed from 2021-23, for the population and various strata. It first attempts to replicate the 2014 IFS methodology applied to the 2021-22 differential (Boileau, O'Brien and Zaranko, 2022; Cribb, Emmerson and Sibieta, 2014). This validates the study, giving it a more credible basis to extend the research to subgroups and the 2022-23 financial year.

The differential is a factor in determining levels of recruitment and retention for the public sector. By highlighting where the gap most favours private sector workers, government can make informed decisions on how to allocate funds for annual pay awards to attract and retain skilled staff. The other key factor in determining recruitment and retention is pensions, which this study does not investigate.

"The public sector pay bill amounted to £233 billion in 2021-22, representing more than 20% of total government spending, 33% of what government spends on public services and almost 10% of national income" (Boileau, O'Brien and Zaranko, 2022). Thus, decisions on pay awards are important in a stretched government budget and debt standing at around 94% of GDP (ONS, 2026a).

The study covers a tumultuous time with numerous notable external shocks influencing the differential. It includes the end of the global pandemic in 2021-22 and the last two quarters of the furlough scheme. By extending analysis to 2022-23, the period coincides with the wave of public sector strikes and the cost of living crisis, during which public sector pay awards lagged behind double-digit inflation. These shocks are unlikely to be uniform, impacting different groups' and sectors' wages differently. Thus, studying the wage differential helps us gain insight into the nature of this asymmetry.

The questions the study set out to explore were:

- Can the IFS (2022) estimates for 2021-22 be replicated?
- How did the differential change in 2022-23?
- How does it vary by sex, region and qualification?

3. Data and methodology

The data used is Labour Force Survey data spanning 2021q2 to 2023q1 (ONS, 2025a–c; 2026b–f). The data is recorded in waves, with only waves 1 and 5 answering the income questions, thus income weights (PIWT22) were used which are calculated based only on these relevant quarters.

Table 1: Sample sizes

	Private sector	Public sector	Total
2021-22	19,575	9,242	28,817
2022-23	15,252	7,146	22,398

Unweighted respondents: employees aged 20–59 in LFS waves 1 and 5 with valid hourly pay, after trimming.

The sample was then trimmed to include only employees between 20-59, and all observations with missing value for any variables included in the regressions were dropped. The sample only includes employees, excluding the self-employed as income data is not collected. For each financial year, the bottom percentile of hourly pay was dropped, and the top percentile of hourly pay was dropped provided hours were less than ten. These are weighted percentiles, using PIWT22. These decisions reflect the 2014 IFS methodology, which this study assumed is also used in the 2022 paper. The asymmetry in bottom and top percentile trimming rules is due to top percentiles (£53.42 in 2021-22) most likely reflecting an error if hours are

underreported. Bottom percentile wages (£3.85) are far below the 2021-22 minimum wage and thus is likely to be an error regardless of hours. The model for the differential uses an OLS regression, as seen in equation 1.

Equation 1: ln (hourly pay) = $\alpha + \beta \cdot$public + controls + error

Equation 2: $(e^{\beta} - 1) \times 100$ (Halvorsen, and Palmquist,1980)

First, ln (hourly pay) is chosen so that the coefficient on the public sector dummy variable can easily be converted into a percentage differential using the formula in equation 2.

Second, public represents a public sector dummy variable, which takes the value of one when an employee works in the public sector. The raw differential's is quarterly adjusted with no other controls. A positive differential means that public sector workers are paid more on average.

Third, the set of controls for the conditional differential are age and age², experience and experience² interacted with three education groups, a seven-category education, region and quarter, all interacted with sex. Sex is interacted with all other variables to allow the effect of the other controls to vary between sexes. Age and experience are interacted with themselves to allow for a non-linear relationship with wages. It is necessary for quarterly adjustment to stop quarterly changes, that influence both public and private sector wages, from impacting the differential due to random samples changing between quarters. Experience is also interacted with education to allow for the impact of experience on wages to vary between workers with different qualifications.

All standard errors reported are robust and so the violation of the homoskedasticity assumption in the OLS is not a problem. All confidence intervals are calculated using the delta method which approximates the standard error of a non-linear function using a first order Taylor approximation evaluated at the coefficient. Thus, all confidence intervals are symmetric. This is an approximation, but it is a good one with such a large sample size. The confidence intervals are all at the 95% confidence level (5% significance level).

For sub-group analyses based on sex, education and region, the public dummy was interacted with each variable respectively (one at a time, in separate regressions). This allowed the coefficient on the public sector dummy variable to vary between each sub-group.

A sensitivity test is also carried out on the choice of education variable, as it was unclear in the papers whether the IFS distinguished between lower and higher degree levels. The reported conditional differential uses the finer variable that separates the two qualifications, yet the regression is repeated using the coarser variable to investigate whether this judgement call influences the results.

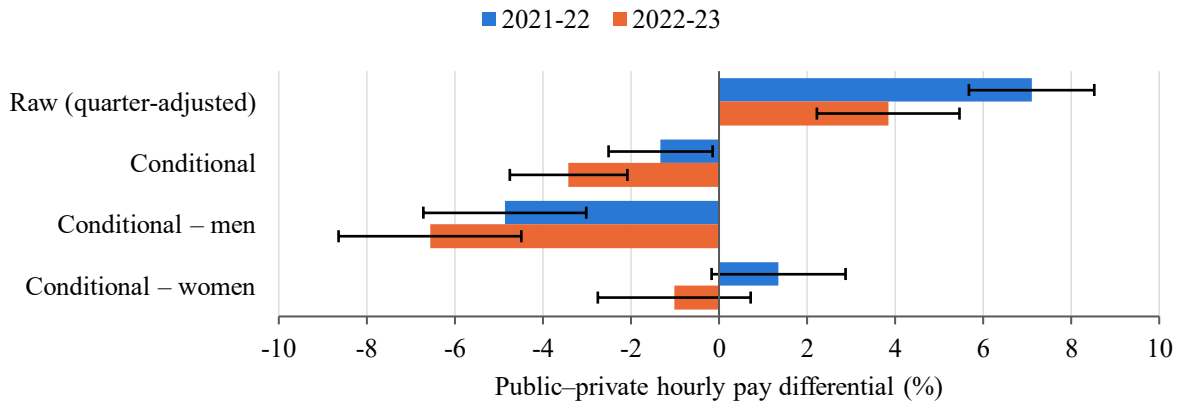
4. Results

Raw differential

The raw differential was 7.1% in 2021-22 (figure 1). In 2022-23 the raw differential fell to 3.8%, favouring public sector workers less than the previous financial year (the 95% CIs do not overlap, thus a significant change). Both of these results are significant with p-values of

<0.001for both years. This indicates that public sector workers are paid more on average before controlling for background. The IFS study finds a raw differential for 2021-22 was 7%, within the CI of [5.7, 8.5]. This indicates that the types of employees that enter the public sector tend to be paid more, likely due to being older and more qualified on average. However, this is insufficient evidence to conclude that public sector employees with the same background are paid more, reflected by the conditional differential.

Figure 1: Public–private pay differential: raw, conditional and by sex



Source: Labour Force Survey, author's calculations. Conditional estimates control for age, experience, education, region and quarter, all interacted with sex. Bars show the percentage differential; lines show 95% confidence intervals. Positive = public sector pays more.

Conditional differential

The conditional differential was -1.3% in 2021-22, falling to -3.4% in 2022-23 (figure 1). The IFS figure for 2021-22 was-0.7%, lying within this studies CI of [-2.5, -0.1]. Both of these figures are significant, with p-values of 0.028 and <0.001 respectively, and a standard z-test shows that the widening of the discount is also significant. This suggests that employees of the same background are paid less in the public sector. Boileau, O'Brien and Zaranko note that this is 'now less favourable to the public sector than at any point in the past 30 years' (2022).

A sensitivity test was carried out on these results to investigate whether the judgement to use a finer education variable influenced the results significantly. The study repeats the controlled regression using an education variable that combined undergraduate and postgraduate degrees into a degree or higher qualification level. This regression produced a conditional differential of -0.7% in 2021-22, which was not statistically different from zero at the 5% significance level. Interestingly, this matches the figure produced by the IFS in 2022 suggesting that they may have used this coarser qualification variable. In contrast, the 2022-23 conditional differential, when education is accounted for using the coarser variable was -2.8%, a significant result with a p-value of <0.001. Thus, unlike 2021-22, the 2022-23 conditional pay differential is robust to the choice of education variable.

Conditional differential by sex

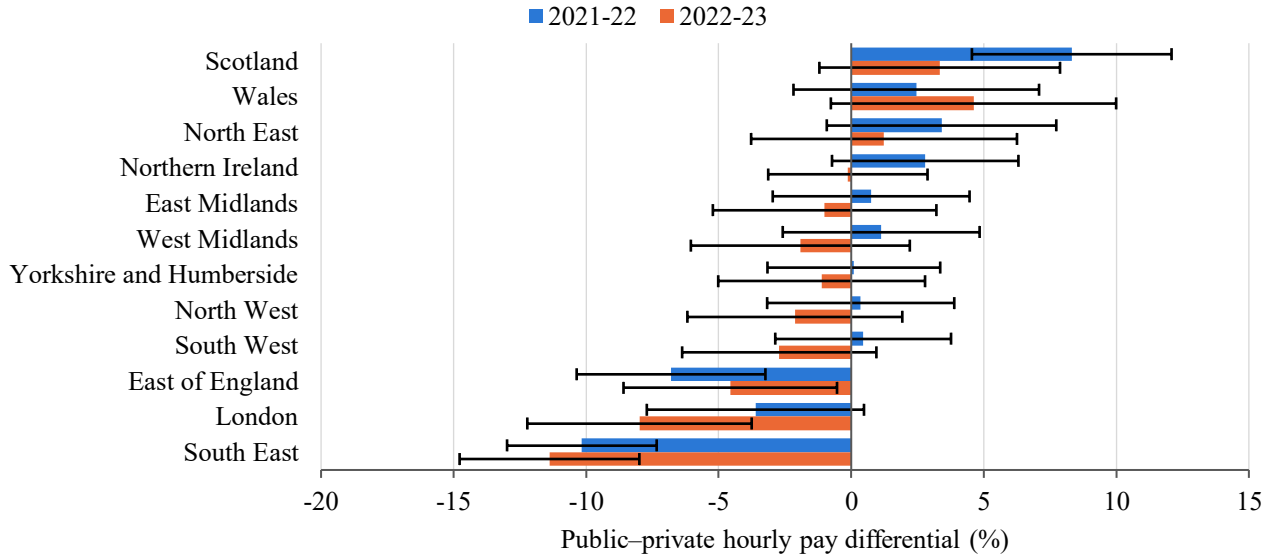
The conditional differential for women was 1.4% in 2021-22 and -1% in 2022-23, with neither of these results being significant. A conditional discount of -4.9% and -6.6% in 2021-22 and 2022-23 respectively was found for men, both of which were significant with p-values

of <0.001. The 2021-22 values found by the IFS were 2.2% and -4.6% for women and men respectively. IFS figures lie in the corresponding confidence intervals of this study. This suggests that male public sector employees are paid less than their private sector counterparts whilst the gap amongst female public employees is indistinguishable from zero. This could be a result of females working in lower paid jobs on average, and the public sector pay scale is more compressed than the range of private wages.

Overall, none of the study's findings contradict the IFS's numbers for 2021-22, thus the methodology used has a credible basis for extension to the next financial year and across various different sub-groups.

Conditional differential by region

Figure 2: By region



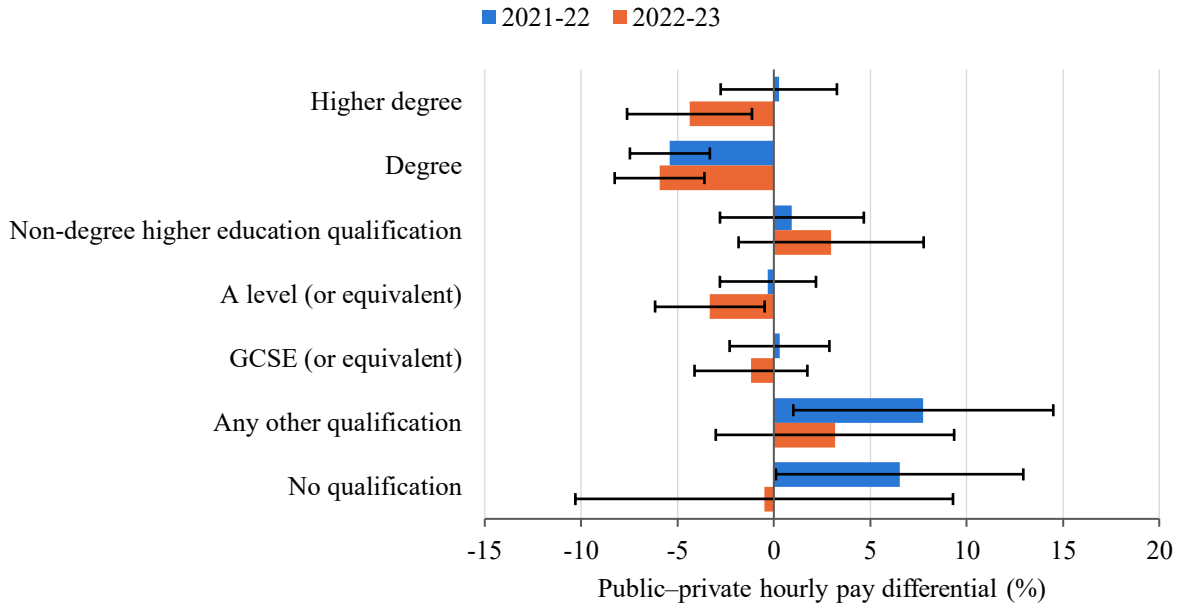
Source: Labour Force Survey, author's calculations. Conditional estimates control for age, experience, education, region and quarter, interacted with sex. Bars show the percentage differential; lines show 95% confidence intervals. Positive = public sector pays more.

The regional analysis helps capture just how focused the conditional wage differential is in certain areas of the UK. There were only three significant results in 2021-22, Scotland with a pay gap of 8.3%, the East of England at -6.8% and the South East with a gap of -10.2%, all with p-values <0.001. This, in part, is due to the far larger confidence intervals associated with much smaller sample sizes. In Scotland, the public premium could reflect the more generous pay awards conceded by the Scottish government. The general trend is that the more affluent the area of the UK, the less favourable the pay differential is to public sector workers. This is likely a result of the compressed national public sector pay scale, which is less sensitive to region than private sector wages. Thus, in poorer regions public sector wages are relatively high, whilst in more affluent areas they are relatively low. (Boileau, O'Brien and Zaranko, 2022)

In 2022-23 the premium in Scotland was no longer significant, whereas the London public sector discount reached -8%, which is statistically different from zero with a p-value of <0.001. The pay differentials in the South East and East of England remained significant, with the East's discount falling to 4.6% and the discount in the South East growing to 11.4%.

Conditional public–private pay differential by highest qualification

Figure 3: By highest qualification



Source: Labour Force Survey, author's calculations. Conditional estimates control for age, experience, education, region and quarter, interacted with sex. Bars show the percentage differential; lines show 95% confidence intervals. Positive = public sector pays more.

The pay differential also varies dramatically by qualification. Undergraduate degree holders in the public sector faced a 5.4% discount in 2021-22 and 5.9% in 2022-23, both statistically significant with p values <0.001. The pay differential for postgraduate degree holders was insignificant in 2021-22 at 0.3%, yet by 2022-23 they faced a significant 4.4% discount, with a value of 0.008. In 2022-23 the discount for A level holders is significant at 3.3%, with a p-value of 0.022. However, these discounts are not seen amongst less qualified workers. Public sector employees with no qualification benefit from a 6.5% premium for working in the public sector in 2021-22, significant with a p value of 0.046. It is worthing noting that this is only just about significant, and the confidence interval spans nearly 13% due to a relatively small sample size of 795. By 2022-23 the gap for those with no qualification is insignificant. Similar to the regional analysis, this may be due to the compressed pay scale, and the fact that lower qualified jobs tend to be lower paid, whilst undergraduate and postgraduate holders will be paid relatively more. However, this logic would suggest that the discount for post-graduate holders would be larger than that of undergraduate holders, yet the opposite is true (CI's do not overlap in 2021-22). This inconsistency may be due to more public sector jobs listing post-graduate degrees as a pre-requisite, and thus public sector employers price in premiums for these jobs.

5. Limitations

Furlough

One limitation is that the furlough quarters (2021q2 and 2021q3) are not adjusted for in the study. There is a possibility that the furlough scheme impacted private and public sector wages differently and thus caused a change in the pay differential. It is also debatable whether furlough should be controlled for. On one hand, if it did cause a change in the differential then that change reflects a tangible change in the conditional pay gap, whereas our other controls correct sampling issues (quarterly adjustment) and control for

characteristics. On the other hand, if the government are using this data to make decisions on longer term public sector pay awards, then a temporary factor such as furlough should be controlled for.

The crude test carried out by the study (appendix Table 2), uses a difference in differences approach to see whether the furlough quarters were associated with a change in the differential. The wage differentials for the furlough and non-furlough quarters of 2021-22 were found, and the difference between them was recorded. This difference was compared to the difference between the differentials in 2022q2-3 and 2022q4-2023q1, to investigate whether the change associated with furlough was actually a result of regular seasonal trends. These calculations indicate that the furlough quarters were about 2.4 points more favourable to public sector workers than if they followed the same seasonal pattern as 2022-3. However, the results are insignificant with a large standard error and p value of 0.18 (appendix calculations 1).

There are numerous issues with this test:

1. The control for seasonal adjustment takes 2022-23 as representative of a normal financial year. Due to shocks such as covid, cost of living, public sector strikes and geopolitical conflicts likely having varying impacts on each financial year, it is problematic to call 2022-23 a 'placebo'
2. Furthermore, any good control for seasonal trends would use a sample spanning more than one financial year
3. The independence assumption required for these crude difference in differences approach breaks down, because LFS respondents answer the income questions in waves 1 and 5. Thus the incomes reported in 2021-22 and 2022-23 have a positive correlation with each other. However, this would mean true standard errors are smaller than what has been reported and so the test could be more sensitive

An extension of this study could compile LFS data over a far longer period to create a seasonal control. Then, create a dummy variable taking the value of 1 for 2021q2-3 and add this variable to the controls.

Pensions

This study only looks at wages, neglecting pensions completely. For a more complete picture of differences in remuneration for public and private sector employees, one must also study differences in pensions, which tend to be higher for public sector employees.

Experience variable

The experience variable is not a perfect control because it is measures the maximum possible experience a worker could have attained since finished their highest qualification. It assumes that workers had no career breaks and no experience prior to their highest qualification. The implication of this depends on whether these exceptions are more concentrated in one of the two sectors.

Causality and association

These pay differentials are not causal as the controls do not exhaust all of the characteristics that could determine a workers wages unrelated to the public sector dummy variable (productivity of workers, preferences etc). Thus, this study answers the question of how wages differ between two workers with comparable backgrounds in the public and private sector rather than answering how wages would differ for an individual person in either sector.

Factors that are related to the public sector dummy variable such as occupation are correctly not controlled since this strips away part of the pay differential caused by the decision to work in the public or private sectors.

To answer the latter question, one could identify individuals who switched sectors between waves 1 and 5 of the survey and track the impact on their wages. Alternatively, the use of an instrumental variable that correlates with the public sector dummy but not with the wages other than its association with the public sector could be used to assert that the differential is caused by choice of sector. However, to carry out such test well, richer data and more time would be necessary.

Other assumptions

The methodology also assumes that the 2022 IFS study uses the same methodology as Cribb, Emmerson and Sibieta (2014). This assumes that the sample size is restricted to 20-59 year olds in 2022, which could not be verified.

It is also assumed that between 2021-22 and 2022-23 there is no household clustering. In reality, in the LFS survey every adult member of a household is surveyed, yet the standard errors are calculated under the assumption that observations for pay, and other variables are independent of each other. Thus, this assumption breaks down, and standard errors are likely to be underreported slightly. This could undermine the results that are only just significant, and so an extension of this work would be calculating clustered standard errors.

AI use

The project's purpose was to improve my understanding of econometrics and my competency in Stata. Thus, Claude by Anthropic was used to learn how to code steps of the preparation, cleaning and analysis. AI was often used to debug the code when it returned error. Comments on the Stata code are mine, where I attempt to justify judgement calls, explain what lines of code do, and why they are necessary. AI was used in cases to help me understand the code and thus informed some of my comments. AI was also used to write the python scripts to create the tables and figures in this report. Results from the regressions were uploaded to Claude, which converted them into figures and tables, to save time. AI also reviewed drafts of my write-up to catch errors. No raw data was uploaded to AI during this study.

6. Conclusions

Overall, a few key trends are worth noting.

1. The raw pay differential remains favourable to public sector workers

2. The public-private sector conditional pay differential varies dramatically between sexes, regions and qualification levels
3. Subgroups associated with lower incomes tend to be higher paid in the public sector relative to private sector counterparts
4. Between 2021-22 and 2022-23 the conditional pay differential amongst most subgroups became less favourable to the public sector

The study also replicated the IFS 2022 figures using the IFS 2014 methodology to, at least, the 95% confidence level establishing a credible basis for the extension of the study to 2022-23 and more granular analysis.

Key takeaways for policy include:

1. Men, degree holders, and workers from London and the south of England face the largest discount, thus recruitment and retention among these sub-groups may be difficult
2. Compressed national pay scales are the likely cause of such a range in premiums and discounts for subgroups. There may be a case for more regional flexibility for recruitment and retention. This, of course, may have adverse impacts on regional disparities given that the discount is larger in more affluent areas. Policy recommendations could consider other policies for redistribution, whilst closing the differential for recruitment and retention
3. The discount widened significantly in 2022-23 consistent with the cost of living crisis and insufficient pay awards. This suggests that the differential should be of increasing importance to policymakers

7. Appendix

Sample

	Private sector	Public sector	Total
2021-22	19,575	9,242	28,817
2022-23	15,252	7,146	22,398

Unweighted respondents: employees aged 20–59 in LFS waves 1 and 5 with valid hourly pay, after trimming.

Table 1. Headline estimates

	2021-22	95% CI	p	N	2022-23	95% CI	p	N
Raw (quarter-adjusted)	7.1***	[5.7, 8.5]	<0.001	28,817	3.8***	[2.2, 5.5]	<0.001	22,398
Conditional	-1.3**	[-2.5, -0.1]	0.028	28,817	-3.4***	[-4.8, -2.1]	<0.001	22,398
Conditional (HIQUL15D education controls)	-0.7	[-1.9, 0.5]	0.271	28,817	-2.8***	[-4.1, -1.4]	<0.001	22,398
Conditional – men	-4.9***	[-6.7, -3.0]	<0.001	13,265	-6.6***	[-8.6, -4.5]	<0.001	10,371

Conditional – women	1.4*	[-0.2, 2.9]	0.081	15,552	-1.0	[-2.8, 0.7]	0.251	12,027

The HIQUL15D row is a sensitivity check: degree and higher degree are combined in the education controls. *Percentage differential, 100×(exp(b)−1), weighted by PIWT22 with robust standard errors. p-values test whether each differential differs from zero; yellow = significant at the 5% level. *** p<0.01, ** p<0.05, * p<0.1.*

Table 2. Furlough check (conditional differential)

	2021-22	95% CI	p	N	2022-23	95% CI	p	N
Oct–Mar quarters (no furlough)	-1.7*	[-3.4, 0.0]	0.052	13,887	-2.5**	[-4.5, -0.5]	0.014	10,416
Apr–Sep quarters (furlough in 2021-22)	-1.0	[-2.7, 0.6]	0.216	14,930	-4.3***	[-6.0, -2.5]	<0.001	11,982

2021-22: Apr–Sep quarters were under the furlough scheme. 2022-23 is a placebo using the same calendar quarters. *Percentage differential, 100×(exp(b)−1), weighted by PIWT22 with robust standard errors. p-values test whether each differential differs from zero; yellow = significant at the 5% level. *** p<0.01, ** p<0.05, * p<0.1.*

Calculation 1: Difference-in-differences of the conditional differential: (b − a) − (d − c) = 2.43 points; SE = $\sqrt{\Sigma SE^2}$ = 1.82, assuming independent estimates; z = 1.34, two-sided p = 0.18 where a and b are the 2021-22 Oct–Mar and Apr–Sep estimates, and c and d the same for 2022-23 (Table 2).

Table A1. By region

	2021-22	95% CI	p	N	2022-23	95% CI	p	N
Scotland	8.3***	[4.6, 12.1]	<0.001	2,019	3.3	[-1.2, 7.9]	0.150	1,547
Wales	2.5	[-2.2, 7.1]	0.299	1,390	4.6*	[-0.8, 10.0]	0.093	940
North East	3.4	[-0.9, 7.7]	0.123	1,352	1.2	[-3.8, 6.3]	0.628	1,005
Northern Ireland	2.8	[-0.7, 6.3]	0.120	2,097	-0.1	[-3.1, 2.9]	0.934	2,352
East Midlands	0.8	[-3.0, 4.5]	0.692	2,222	-1.0	[-5.2, 3.2]	0.641	1,832
West Midlands	1.1	[-2.6, 4.8]	0.551	2,281	-1.9	[-6.1, 2.2]	0.362	1,732
Yorkshire and Humberside	0.1	[-3.2, 3.4]	0.953	2,474	-1.1	[-5.0, 2.8]	0.575	1,903
North West	0.4	[-3.2, 3.9]	0.843	2,825	-2.1	[-6.2, 1.9]	0.304	2,082

South West	0.4	[-2.9, 3.8]	0.791	2,848	-2.7	[-6.4, 0.9]	0.146	2,117
East of England	-6.8***	[-10.4, -3.2]	<0.001	2,617	-4.6**	[-8.6, -0.5]	0.026	2,169
London	-3.6*	[-7.7, 0.5]	0.084	2,607	-8.0***	[-12.2, -3.8]	<0.001	1,787
South East	-10.2***	[-13.0, -7.3]	<0.001	4,085	-11.4***	[-14.8, -8.0]	<0.001	2,932

*Percentage differential, 100×(exp(b)−1), weighted by PIWT22 with robust standard errors. p-values test whether each differential differs from zero; yellow = significant at the 5% level. *** p<0.01, ** p<0.05, * p<0.1.*

Table A2. By highest qualification

	2021-22	95% CI	p	N	2022-23	95% CI	p	N
Higher degree	0.3	[-2.8, 3.3]	0.867	4,787	-4.4***	[-7.6, -1.1]	0.008	3,849
Degree	-5.4***	[-7.5, -3.3]	<0.001	8,579	-5.9***	[-8.3, -3.6]	<0.001	6,679
Non-degree higher education qualification	0.9	[-2.8, 4.7]	0.622	2,522	3.0	[-1.8, 7.8]	0.225	1,782
A level (or equivalent)	-0.3	[-2.8, 2.2]	0.810	6,017	-3.3**	[-6.2, -0.5]	0.022	4,763
GCSE (or equivalent)	0.3	[-2.3, 2.9]	0.824	4,791	-1.2	[-4.1, 1.7]	0.427	3,526
Any other qualification	7.8**	[1.0, 14.5]	0.024	1,326	3.2	[-3.0, 9.4]	0.315	1,129
No qualification	6.5**	[0.1, 12.9]	0.046	795	-0.5	[-10.3, 9.3]	0.920	670

*Percentage differential, 100×(exp(b)−1), weighted by PIWT22 with robust standard errors. p-values test whether each differential differs from zero; yellow = significant at the 5% level. *** p<0.01, ** p<0.05, * p<0.1.*

Stata code can be found on GitHub at: https://github.com/laban-davies/Replicating-and-extending-the-IFS-2022-study-on-the-public-private-sector-differential

8. References

Boileau, B., O'Brien, L. and Zaranko, B. (2022) 'Public spending, pay and pensions', in *IFS Green Budget 2022*. IFS Report R219. London: Institute for Fiscal Studies. Available at: https://ifs.org.uk/sites/default/files/2022-10/Public-spending-pay-and-pensions-R219.pdf (Accessed: 10 October 2026).

Cribb, J., Emmerson, C. and Sibieta, L. (2014) *Public sector pay in the UK*. IFS Report R97. London: Institute for Fiscal Studies. Available at: https://ifs.org.uk/sites/default/files/output_url_files/r97.pdf (Accessed: 10 October 2026).

Halvorsen, R. and Palmquist, R. (1980) 'The interpretation of dummy variables in semilogarithmic equations', *American Economic Review*, 70(3), pp. 474–475.

Office for National Statistics (ONS) (2025a) *Quarterly Labour Force Survey, January–March 2023* [data collection]. 4th edn. UK Data Service. SN: 9097. Available at: http://doi.org/10.5255/UKDA-SN-9097-4 (Accessed: 5 October 2026).

Office for National Statistics (ONS) (2025b) *Quarterly Labour Force Survey, July–September 2022* [data collection]. 6th edn. UK Data Service. SN: 9027. Available at: http://doi.org/10.5255/UKDA-SN-9027-6 (Accessed: 5 October 2026).

Office for National Statistics (ONS) (2025c) *Quarterly Labour Force Survey, October–December 2022* [data collection]. 4th edn. UK Data Service. SN: 9052. Available at: http://doi.org/10.5255/UKDA-SN-9052-4 (Accessed: 5 October 2026).

Office for National Statistics (ONS) (2026a) *Public sector finances, UK: August 2026*. Released 22 September 2026. Available at: https://www.ons.gov.uk/economy/governmentpublicsectorandtaxes/publicsectorfinance/bulletins/publicsectorfinances/august2026 (Accessed: 10 October 2026).

Office for National Statistics (ONS) (2026b) *Quarterly Labour Force Survey, April–June 2021* [data collection]. 5th edn. UK Data Service. SN: 8826. Available at: http://doi.org/10.5255/UKDA-SN-8826-5 (Accessed: 1 October 2026).

Office for National Statistics (ONS) (2026c) *Quarterly Labour Force Survey, April–June 2022* [data collection]. 5th edn. UK Data Service. SN: 8999. Available at: http://doi.org/10.5255/UKDA-SN-8999-5 (Accessed: 5 October 2026).

Office for National Statistics (ONS) (2026d) *Quarterly Labour Force Survey, January–March 2022* [data collection]. 6th edn. UK Data Service. SN: 8957. Available at: http://doi.org/10.5255/UKDA-SN-8957-6 (Accessed: 1 October 2026).

Office for National Statistics (ONS) (2026e) *Quarterly Labour Force Survey, July–September 2021* [data collection]. 7th edn. UK Data Service. SN: 8872. Available at: http://doi.org/10.5255/UKDA-SN-8872-7 (Accessed: 1 October 2026).

Office for National Statistics (ONS) (2026f) *Quarterly Labour Force Survey, October–December 2021* [data collection]. 6th edn. UK Data Service. SN: 8915. Available at: http://doi.org/10.5255/UKDA-SN-8915-6 (Accessed: 1 October 2026).